MACSE603 — LABORATORY RECORD

Visual Analytics Lab

Assessment 1 & Assessment 2 — Power BI Sales Analytics

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Name: Vedant Nimkar | **Registration No.:** 26MML0045
Subject: Visual Analytics Lab (MACSE603) | **Programme:** M.Tech Computer Science and Engineering
Tools: Power BI Desktop, SQL Server, Power Query (M), DAX, Python

Aim

Assessment 1 (QP1) — set up the environment and connect to a data source, apply data cleansing and transformation in Power Query, and design a data model with relationships, column properties and hierarchies using a sales dataset.

Assessment 2 (QP2) — create calculated tables, calculated columns and measures using DAX; use `CALCULATE()` to manipulate filter context and Time Intelligence functions for sales analysis; then design and publish a report with basic visualizations.

Dataset

A retail sales dataset was generated in a star-schema layout. The fact table was deliberately produced in a *dirty* state so that the cleansing requirements of QP1 Step 2 operate on genuine data-quality defects.

Table	Rows	Role
`FactSales_RAW`	6,012	Fact — order lines (uncleansed)
`DimProduct`	22	Product, subcategory, category, pricing
`DimCustomer`	220	Customer name, segment, territory
`DimTerritory`	5	Region, country, group
`DimDate`	1,096	Calendar 2023-01-01 to 2025-12-31

Files are in `data/` as CSV and as a single `SalesData.xlsx` workbook. `sql/01_create_load.sql` recreates the same tables in SQL Server.

Assessment 1 (QP1)

Step 1 — Setting up the environment and connecting to the data source

Procedure

1. Install **Power BI Desktop** (Microsoft Store or standalone installer) and **SQL Server** with SQL Server Management Studio.
2. Execute `sql/01_create_load.sql` in SSMS. This creates the `VisualAnalyticsLab` database and loads all five tables.
3. In Power BI Desktop choose **Home → Get data → SQL Server**.
4. Enter server `localhost` (or `.\SQLEXPRESS`), database `VisualAnalyticsLab`, connectivity mode **Import**.
5. In the Navigator select `FactSales`, `DimProduct`, `DimCustomer`, `DimTerritory`, `DimDate` and preview each one.
6. Click **Transform Data** to open Power Query Editor rather than loading immediately.

> If SQL Server is unavailable, use **Get data → Text/CSV** on the files in `data/`, or **Excel workbook** on `SalesData.xlsx`. The remaining steps are identical; only the `Source` line of each query changes.

Step 1 - Raw FactSales preview as loaded from the source

lesOrderNumb	OrderDateText	ProductKey	CustomerKey	TerritoryKey	OrderQuantity	UnitPriceText	DiscountPct	SalesAmount	TotalCost	OrderStatus	Notes
SO50000	11-02-2023	4	6	5	6.0	Rs. 696.13	0.0	4176.78	2580	Shipped	(blank)
SO50001	2025-08-24	11	37	3	8.0	Rs. 54.35	0.0	434.8	96	Shipped	(blank)
SO50002	2023-02-10	13	173	3	2.0	Rs. 118.43	0.15	201.33	116	Shipped	(blank)
SO50003	2023-04-22	2	168	3	1.0	Rs. 1,890.04	0.0	1890.04	980	cancelled	(blank)
SO50004	2023-02-06	22	203	2	2.0	Rs. 206.83	0.1	372.29	192	Shipped	(blank)
SO50005	2024-04-29	14	19	1	2.0	Rs. 198.50	0.0	397.0	132	shipped	(blank)
SO50006	2025-12-09	5	191	1	3.0	Rs. 286.03	0.0	858.09	435	SHIPPED	(blank)
SO50007	13-07-2025	13	182	4	5.0	Rs. 221.02	0.0	1105.1	290	Shipped	(blank)
SO50008	2023-03-20	3	190	4	5.0	Rs. 514.33	0.0	2571.65	1550	Pending	(blank)
SO50009	2024-10-22	14	4	3	1.0	Rs. 148.03	0.0	148.03	66	Cancelled	(blank)
SO50010	2025-10-24	6	185	4	4.0	Rs. 225.34	0.0	901.36	312	Pending	(blank)
SO50011	26/07/2025	4	107	5	1.0	Rs. 1,213.93	0.05	1153.23	430	shipped	(blank)

Full raw table: 6,012 rows x 12 columns. Note the text dates in mixed formats, currency stored as text, blank Notes column and NaN quantities.

Figure 1 — Raw FactSales as loaded from the source, showing text dates in mixed formats, currency held as text, a blank Notes column and missing quantities

SCREENSHOT SLOT

Paste your Power BI Navigator / data preview screenshot here

Step 2 — Data cleansing and transformation

Data-quality defects identified

Defect	Occurrences	Treatment
Exact duplicate rows	12	`Table.Distinct`
Null `OrderQuantity`	40	rows removed
Null `SalesAmount`	35	recomputed from price × qty × (1 − discount)
Negative `OrderQuantity`	8	rows removed as entry errors
Blank `Notes` column	1 column	column removed
Mixed date formats	3 formats	parsed to a true `date` type
Currency stored as text	all rows	`"Rs. 1,234.00"` → `1234.00`
Untrimmed / inconsistent status	many	`Text.Proper(Text.Trim())`
Cancelled orders	949	excluded from sales analysis

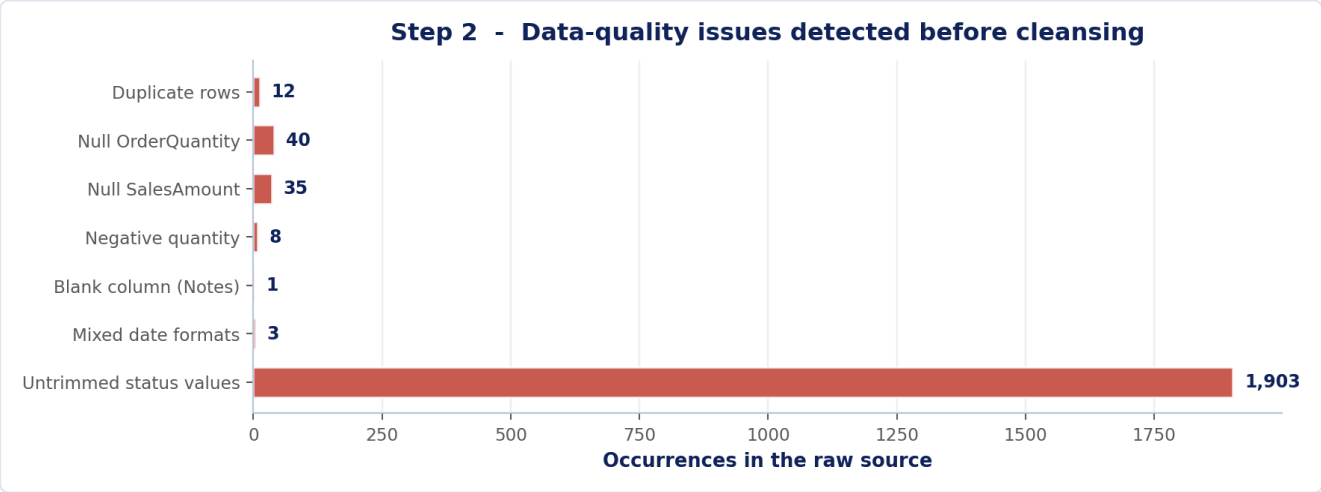


Figure 2 — Data-quality issues detected in the raw source before cleansing

Applied steps and resulting row count

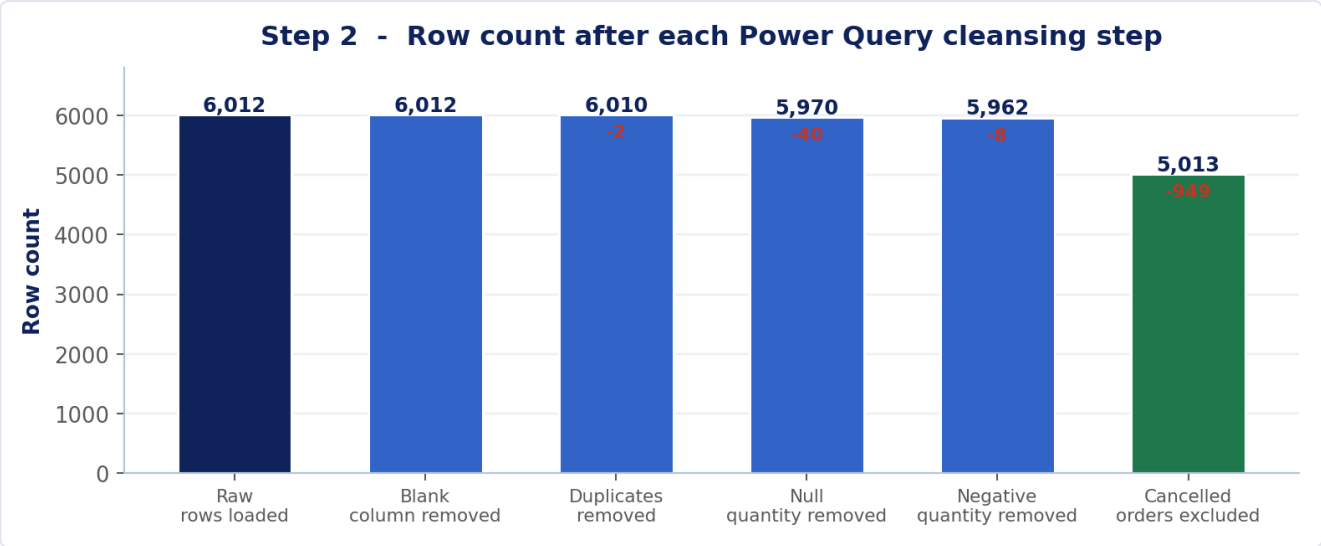


Figure 3 — Row count after each Power Query applied step; 6,012 raw rows reduce to 5,013 analysable rows

Power Query M — FactSales

```
let
    // 1. CONNECT ------------------------------------------------------------
    Source = Csv.Document(
        File.Contents("<PATH>\FactSales_RAW.csv"),
        [Delimiter = ",", Encoding = 65001, QuoteStyle = QuoteStyle.Csv]),
    Promoted = Table.PromoteHeaders(Source, [PromoteAllScalars = true]),

    // 2. REMOVE the always-blank column -----------------------------------
    RemovedBlankCol = Table.RemoveColumns(Promoted, {"Notes"}),

    // 3. REMOVE duplicate rows --------------------------------------------
    RemovedDuplicates = Table.Distinct(RemovedBlankCol),

    // 4. PARSE the mixed-format text date into a real date ------------------
    //    Source has yyyy-MM-dd, dd-MM-yyyy and dd/MM/yyyy mixed together.
    AddedOrderDate = Table.AddColumn(RemovedDuplicates, "OrderDate",
        each let t = Text.Replace([OrderDateText], "/", "-"),
                 p = Text.Split(t, "-")
             in  if Text.Length(p{0}) = 4
                 then #date(Number.FromText(p{0}), Number.FromText(p{1}),
                            Number.FromText(p{2}))
                 else #date(Number.FromText(p{2}), Number.FromText(p{1}),
                            Number.FromText(p{0})),
        type date),

    // 5. STRIP the currency text into a decimal ---------------------------
    AddedUnitPrice = Table.AddColumn(AddedOrderDate, "UnitPrice",
        each Number.FromText(
            Text.Trim(Text.Replace(Text.Replace([UnitPriceText], "Rs.", ""),
                                   ",", ""))),
        type number),

    // 6. CLEAN the messy status text (trim + consistent casing) -------------
    CleanedStatus = Table.TransformColumns(AddedUnitPrice,
        {{"OrderStatus", each Text.Proper(Text.Trim(_)), type text}}),

    // 7. SET data types -----------------------------------------------------
    Typed = Table.TransformColumnTypes(CleanedStatus, {
        {"SalesOrderNumber", type text},
        {"ProductKey", Int64.Type},
        {"CustomerKey", Int64.Type},
        {"TerritoryKey", Int64.Type},
        {"OrderQuantity", Int64.Type},
        {"DiscountPct", type number},
        {"SalesAmount", type number},
        {"TotalCost", type number}}),

    // 8. REMOVE rows with a missing quantity --------------------------------
    RemovedNullQty = Table.SelectRows(Typed,
        each [OrderQuantity] <> null),
```

```
    // 9. REMOVE negative-quantity data-entry errors -------------------------
    RemovedNegative = Table.SelectRows(RemovedNullQty,
        each [OrderQuantity] > 0),

    // 10. FILL the missing SalesAmount from price x qty x (1 - discount) -----
    FixedSalesAmount = Table.ReplaceValue(RemovedNegative,
        each [SalesAmount],
        each if [SalesAmount] = null
             then Number.Round([UnitPrice] * [OrderQuantity]
                               * (1 - [DiscountPct]), 2)
             else [SalesAmount],
        Replacer.ReplaceValue, {"SalesAmount"}),

    // 11. EXCLUDE cancelled orders from the sales analysis --------------------
    RemovedCancelled = Table.SelectRows(FixedSalesAmount,
        each [OrderStatus] <> "Cancelled"),

    // 12. ADD a Profit column -------------------------------------------------
    AddedProfit = Table.AddColumn(RemovedCancelled, "Profit",
        each [SalesAmount] - [TotalCost], type number),

    // 13. ADD the DateKey used to join DimDate --------------------------------
    AddedDateKey = Table.AddColumn(AddedProfit, "DateKey",
        each Date.Year([OrderDate]) * 10000
           + Date.Month([OrderDate]) * 100
           + Date.Day([OrderDate]), Int64.Type),

    // 14. DROP the now-redundant raw text columns -----------------------------
    RemovedRawText = Table.RemoveColumns(AddedDateKey,
        {"OrderDateText", "UnitPriceText"}),

    // 15. REORDER for readability ---------------------------------------------
    Reordered = Table.ReorderColumns(RemovedRawText, {
        "SalesOrderNumber", "OrderDate", "DateKey", "ProductKey",
        "CustomerKey", "TerritoryKey", "OrderQuantity", "UnitPrice",
        "DiscountPct", "SalesAmount", "TotalCost", "Profit", "OrderStatus"})
in
    Reordered
```

The complete set of five queries, including the SQL Server variant, is in `code/Step2_PowerQuery_M.txt`.

Step 2 - Cleansed FactSales after all transformations

alesOrderNumbe	OrderDate	ProductKey	OrderQuantity	UnitPrice	DiscountPct	SalesAmount	TotalCost	Profit	OrderStatus	Category
SO50000	2023-02-11	4	6.0	696.13	0.0	4,176.78	2,580.00	1,596.78	Shipped	Electronics
SO50001	2025-08-24	11	8.0	54.35	0.0	434.80	96.00	338.80	Shipped	Accessories
SO50002	2023-10-02	13	2.0	118.43	0.15	201.33	116.00	85.33	Shipped	Accessories
SO50004	2023-06-02	22	2.0	206.83	0.1	372.29	192.00	180.29	Shipped	Software
SO50005	2024-04-29	14	2.0	198.50	0.0	397.00	132.00	265.00	Shipped	Accessories
SO50006	2025-09-12	5	3.0	286.03	0.0	858.09	435.00	423.09	Shipped	Electronics
SO50007	2025-07-13	13	5.0	221.02	0.0	1,105.10	290.00	815.10	Shipped	Accessories
SO50008	2023-03-20	3	5.0	514.33	0.0	2,571.65	1,550.00	1,021.65	Pending	Electronics
SO50010	2025-10-24	6	4.0	225.34	0.0	901.36	312.00	589.36	Pending	Electronics
SO50011	2025-07-26	4	1.0	1,213.93	0.05	1,153.23	430.00	723.23	Shipped	Electronics
SO50012	2023-05-11	8	7.0	367.74	0.0	2,574.18	1,470.00	1,104.18	Shipped	Electronics
SO50013	2024-10-16	11	8.0	40.06	0.0	320.48	96.00	224.48	Shipped	Accessories

Final table: 5,013 rows. Dates are real date values, currency is numeric, status is normalised, Profit added.

Figure 4 — Cleansed FactSales: real date values, numeric currency, normalised status and a derived Profit column

SCREENSHOT SLOT

Paste your Power Query Editor screenshot showing the Applied Steps pane here

Step 3 — Designing the data model

Relationships

#	One side	Many side	Cardinality	Active
1	`DimDate[DateKey]`	`FactSales[DateKey]`	1:*	Yes
2	`DimProduct[ProductKey]`	`FactSales[ProductKey]`	1:*	Yes
3	`DimCustomer[CustomerKey]`	`FactSales[CustomerKey]`	1:*	Yes
4	`DimTerritory[TerritoryKey]`	`FactSales[TerritoryKey]`	1:*	Yes
5	`DimTerritory[TerritoryKey]`	`DimCustomer[TerritoryKey]`	1:*	**No**

Relationship 5 is left **inactive**. Both `FactSales` and `DimCustomer` carry `TerritoryKey`, creating two paths to `DimTerritory`; Power BI permits only one active path. The inactive relationship is later invoked on demand with `USERELATIONSHIP` in Step 5.

`DimDate` is then marked as a date table via **Table tools → Mark as date table → Date**. This is a prerequisite for every Time Intelligence function used in Step 5.

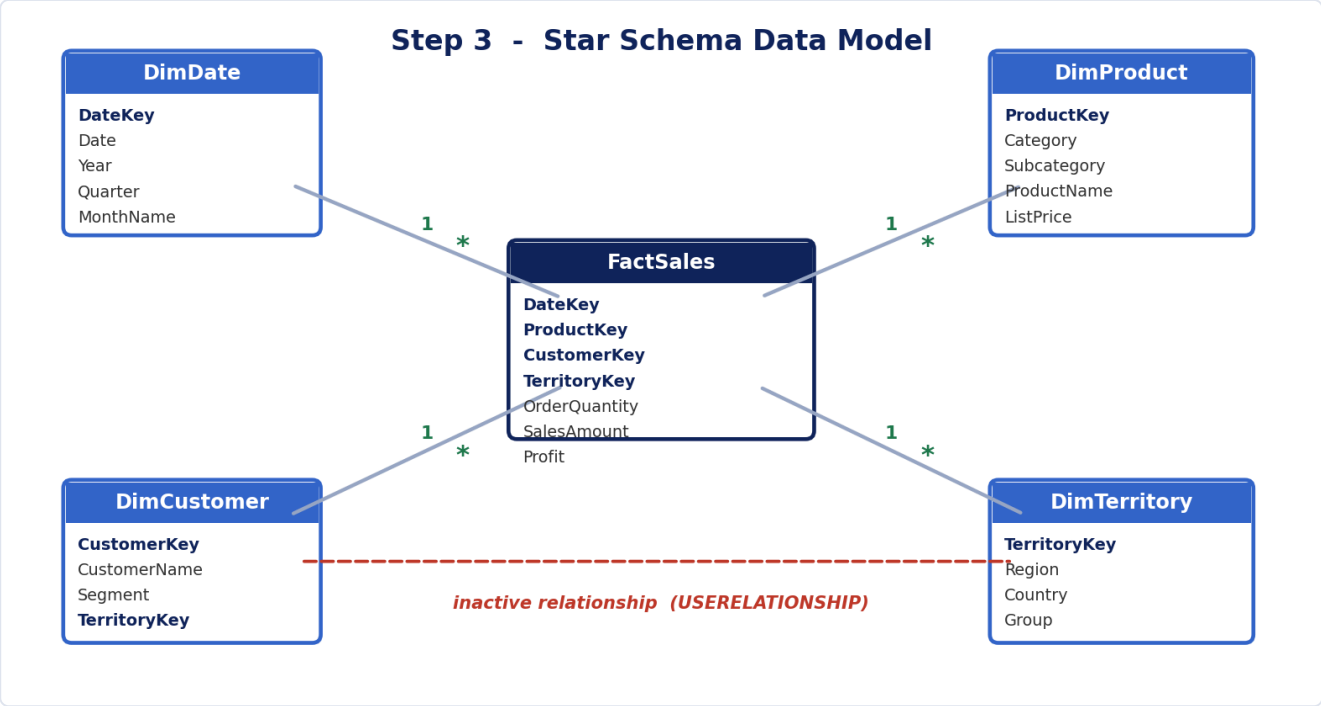


Figure 5 — Star schema: one fact table surrounded by four dimensions, with the inactive Customer→Territory relationship shown dashed

Column properties

Setting	Applied to	Purpose
Hidden	all `*Key` columns	forces report authors to use descriptive attributes
Currency, 2 dp	`SalesAmount`, `TotalCost`, `Profit`, `UnitPrice`	consistent monetary display
Percentage	`DiscountPct`	readable as a rate
Do not summarize	keys, prices, text fields	prevents meaningless auto-aggregation
Sort by column	`MonthName` → `MonthNo`, `Quarter` → `QuarterNo`	chronological rather than alphabetical axis order

Hierarchies

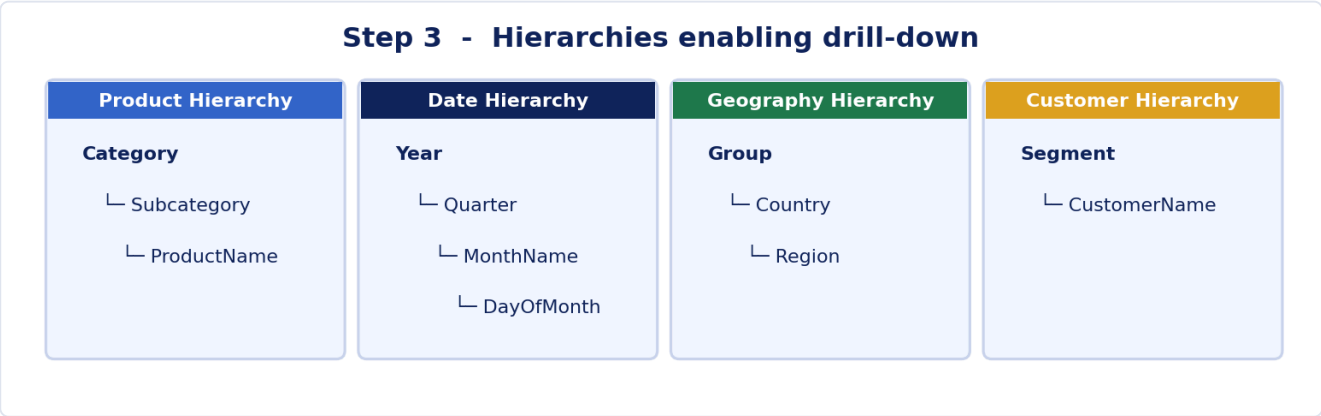


Figure 6 — Four hierarchies created to support drill-down in the Step 6 visuals

SCREENSHOT SLOT

Paste your Power BI Model view screenshot here

Assessment 2 (QP2)

Step 4 — Calculated tables, columns and measures

Calculated columns

```
Net Unit Price = DIVIDE ( FactSales[SalesAmount], FactSales[OrderQuantity] )

Line Margin % =
DIVIDE ( FactSales[SalesAmount] - FactSales[TotalCost],
         FactSales[SalesAmount] )

Order Size Band =
SWITCH ( TRUE (),
    FactSales[OrderQuantity] <= 2, "1-2 units",
    FactSales[OrderQuantity] <= 4, "3-4 units",
    FactSales[OrderQuantity] <= 6, "5-6 units",
    "7+ units" )

Product Category = RELATED ( DimProduct[Category] )

Price Tier =
SWITCH ( TRUE (),
    DimProduct[ListPrice] < 100, "Budget",
    DimProduct[ListPrice] < 500, "Mid-range",
    "Premium" )
```

Core measures

```
Total Sales           = SUM ( FactSales[SalesAmount] )
Total Quantity        = SUM ( FactSales[OrderQuantity] )
Total Cost            = SUM ( FactSales[TotalCost] )
Total Profit          = [Total Sales] - [Total Cost]
Profit Margin %       = DIVIDE ( [Total Profit], [Total Sales] )
Order Count           = DISTINCTCOUNT ( FactSales[SalesOrderNumber] )
Average Order Value   = DIVIDE ( [Total Sales], [Order Count] )
Average Selling Price = DIVIDE ( [Total Sales], [Total Quantity] )
```

Verification — unfiltered model

Measure	Expected value
Total Sales	8,955,541.25
Total Quantity	22,448
Total Cost	4,123,542.00
Total Profit	4,831,999.25
Profit Margin %	53.96%
Order Count	5,008
Average Order Value	1,788.25

Question 1 — top sold product

```
Top Product by Units =
VAR TopProduct =
    TOPN ( 1, ALL ( DimProduct[ProductName] ),
           CALCULATE ( [Total Quantity] ), DESC )
RETURN CONCATENATEX ( TopProduct, DimProduct[ProductName], ", " )

Product Rank by Units =
RANKX ( ALL ( DimProduct[ProductName] ), [Total Quantity], , DESC, DENSE )
```

Result: `GlideMouse Wireless` with **3,204 units**. Ranking by revenue instead returns `GamingBook X17` at **2,534,432.17** — a useful demonstration that unit volume and revenue produce different winners.

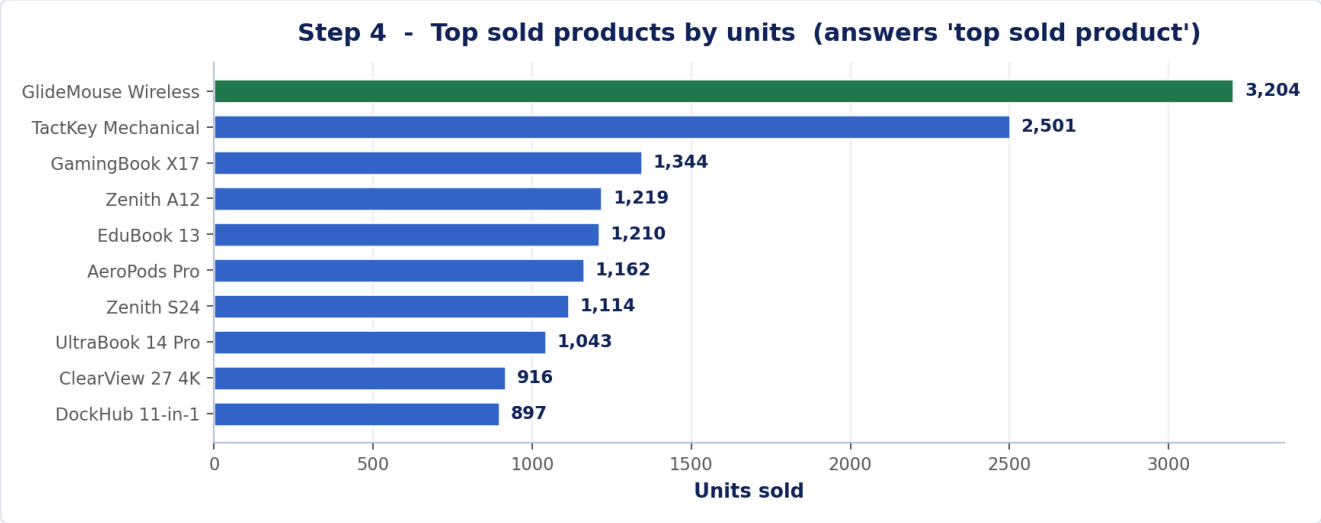


Figure 7 — Top ten products by units sold; the leading product is highlighted

Question 2 — average sales of product category

```
Avg Sales per Order =
DIVIDE ( [Total Sales], DISTINCTCOUNT ( FactSales[SalesOrderNumber] ) )

Avg Sales per Category =
AVERAGEX ( VALUES ( DimProduct[Category] ), [Total Sales] )

Category Share % =
DIVIDE ( [Total Sales], CALCULATE ( [Total Sales], ALL ( DimProduct ) ) )
```

Category	Total Sales	Orders	Avg Sales per Order
Electronics	7,081,992.08	2,172	3,260.59
Furniture	861,784.14	519	1,660.47
Accessories	786,667.35	1,996	394.12
Software	225,097.68	321	701.24

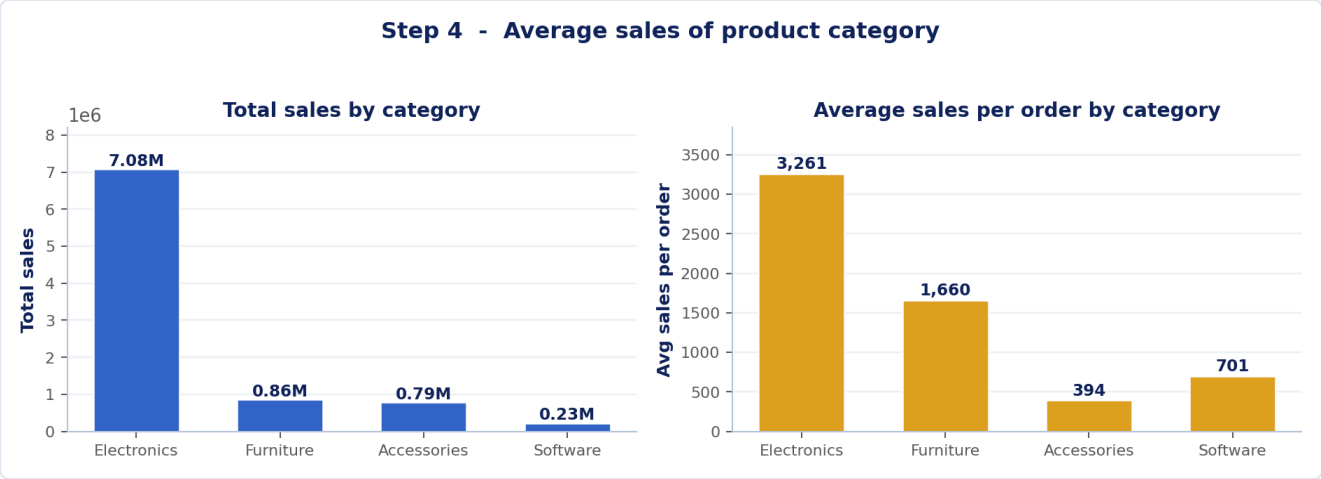


Figure 8 — Total sales and average sales per order by category. Accessories rank second on order count but last on order value

Calculated tables

```
Product Category Summary =
SUMMARIZE ( FactSales, DimProduct[Category],
    "Total Sales",         [Total Sales],
    "Total Quantity",      [Total Quantity],
    "Order Count",         [Order Count],
    "Avg Sales per Order", [Avg Sales per Order] )

Top 5 Products =
TOPN ( 5,
    SUMMARIZE ( FactSales, DimProduct[ProductName],
        "Units Sold", [Total Quantity], "Revenue", [Total Sales] ),
    [Units Sold], DESC )

Date Table DAX =
ADDCOLUMNS ( CALENDAR ( DATE(2023,1,1), DATE(2025,12,31) ),
    "Year", YEAR([Date]), "MonthName", FORMAT([Date],"MMMM"),
    "Quarter", "Q" & QUARTER([Date]) )
```

SCREENSHOT SLOT

Paste your Power BI screenshot of the calculated table and measure results here

Step 5 — Advanced DAX: CALCULATE and Time Intelligence

Manipulating filter context with CALCULATE()

Technique	Function	Effect
Add a filter	`CALCULATE([Total Sales], Category="Electronics")`	narrows the context
Remove all filters	`ALL(DimProduct)`	gives a grand-total denominator
Keep one filter	`ALLEXCEPT(DimProduct, Category)`	share within own category
Respect slicers	`ALLSELECTED(Category)`	share of user selection
Intersect not replace	`KEEPFILTERS(Year=2025)`	preserves an existing Year slicer
Switch relationship	`USERELATIONSHIP(...)`	activates the inactive path from Step 3

```
Category Revenue =
CALCULATE ( [Total Sales], ALLEXCEPT ( DimProduct, DimProduct[Category] ) )

Pct of Category Sales =
DIVIDE ( [Total Sales],
         CALCULATE ( [Total Sales],
                     ALLEXCEPT ( DimProduct, DimProduct[Category] ) ) )

Sales by Customer Territory =
CALCULATE ( [Total Sales],
    USERELATIONSHIP ( DimCustomer[TerritoryKey],
                      DimTerritory[TerritoryKey] ) )
```

Time Intelligence

```
Sales YTD = TOTALYTD ( [Total Sales], DimDate[Date] )
Sales QTD = TOTALQTD ( [Total Sales], DimDate[Date] )

Sales PY  = CALCULATE ( [Total Sales], SAMEPERIODLASTYEAR ( DimDate[Date] ) )

YoY Growth %  = DIVIDE ( [Total Sales] - [Sales PY], [Sales PY] )

Sales Rolling 3M =
CALCULATE ( [Total Sales],
    DATESINPERIOD ( DimDate[Date], MAX ( DimDate[Date] ), -3, MONTH ) )

Sales Cumulative =
CALCULATE ( [Total Sales],
    FILTER ( ALLSELECTED ( DimDate[Date] ),
             DimDate[Date] <= MAX ( DimDate[Date] ) ) )
```

Total sales revenue for each product category, by year

Category	2023	2024	2025
Electronics	2,229,091.18	2,422,281.89	2,430,619.01
Furniture	215,300.29	285,115.53	361,368.32
Accessories	214,422.23	264,027.21	308,217.91
Software	61,936.51	61,233.78	101,927.39

YoY growth trend by product category

Category	2024 vs 2023	2025 vs 2024
Accessories	+23.13%	+16.74%
Electronics	+8.67%	+0.34%
Furniture	+32.43%	+26.74%
Software	−1.13%	+66.45%
Overall	**+11.46%**	**+5.59%**

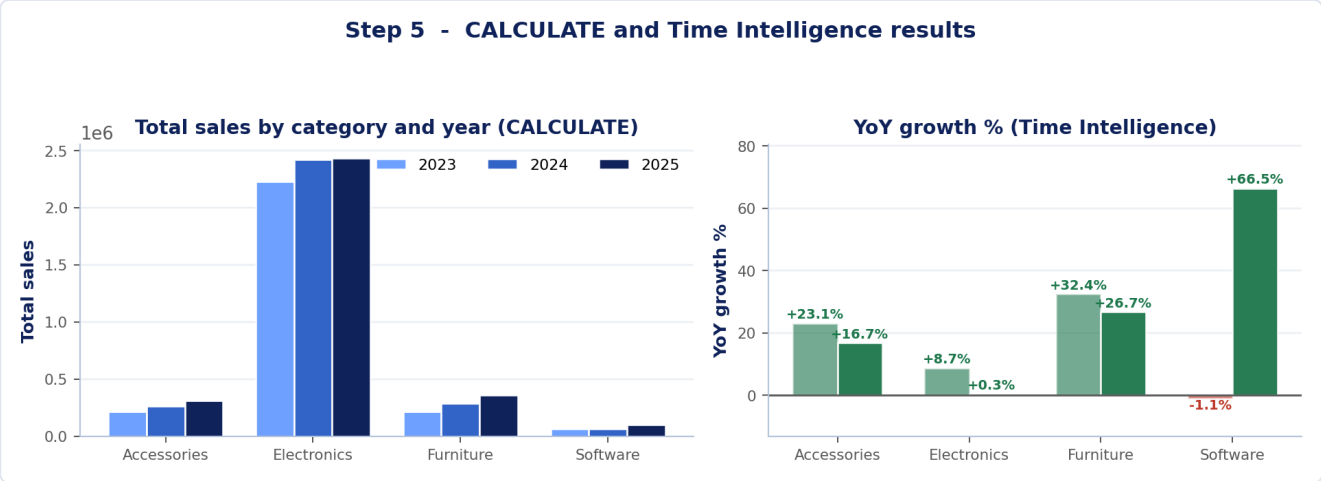


Figure 9 — Category revenue by year produced with CALCULATE, and YoY growth percentage produced with SAMEPERIODLASTYEAR

Electronics dominates revenue but has effectively plateaued in 2025 (+0.34%), while Furniture continues to grow strongly (+26.74%). Software declined in 2024 then rebounded sharply. This divergence is invisible in the headline +5.59% figure and is precisely what the Time Intelligence measures are intended to expose.

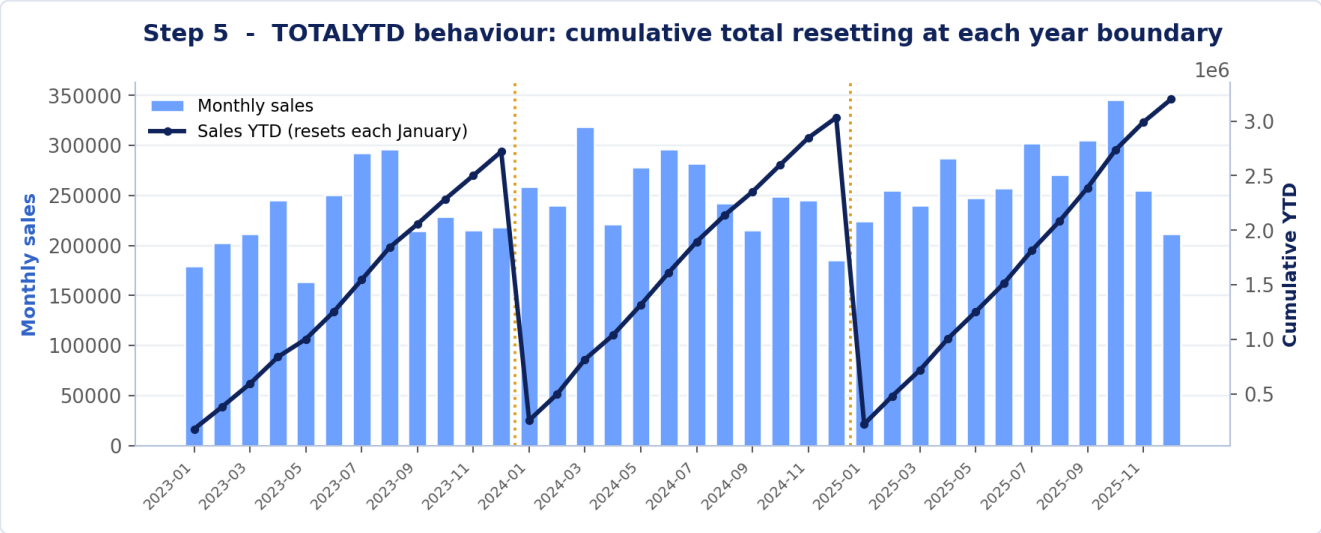


Figure 10 — TOTALYTD behaviour: the cumulative line resets at each year boundary, confirming DimDate is correctly marked as a date table

SCREENSHOT SLOT

Paste your Power BI matrix screenshot showing Total Sales, Sales PY and YoY Growth % here

Step 6 — Designing and publishing the report

Page 1 — Sales Overview

Element	Type	Configuration
KPI row	Cards	Total Sales, Total Quantity, Total Profit, Profit Margin %, YoY Growth %
Sales trend	Line chart	Axis `Date Hierarchy`; values `[Total Sales]`, `[Sales PY]`
Category performance	Stacked column	Axis `Category`, legend `Year`
Top 10 products	Bar chart	Top-N filter 10 by `[Total Quantity]`
Regional split	Map / donut	Location `Region`
Slicers	Year, Category, Region, Segment, OrderStatus	left panel

Page 2 — Category and Time Analysis

Element	Type	Configuration
YoY matrix	Matrix	Rows `Category`, columns `Year`, values `[Total Sales]`, `[Sales PY]`, `[YoY Growth %]` with conditional formatting
Cumulative sales	Area chart	`[Sales Cumulative]`
Rolling average	Line chart	`[Total Sales]` vs `[Sales 3M Moving Avg]`
Monthly YTD	Line and column	column `[Total Sales]`, line `[Sales YTD]`

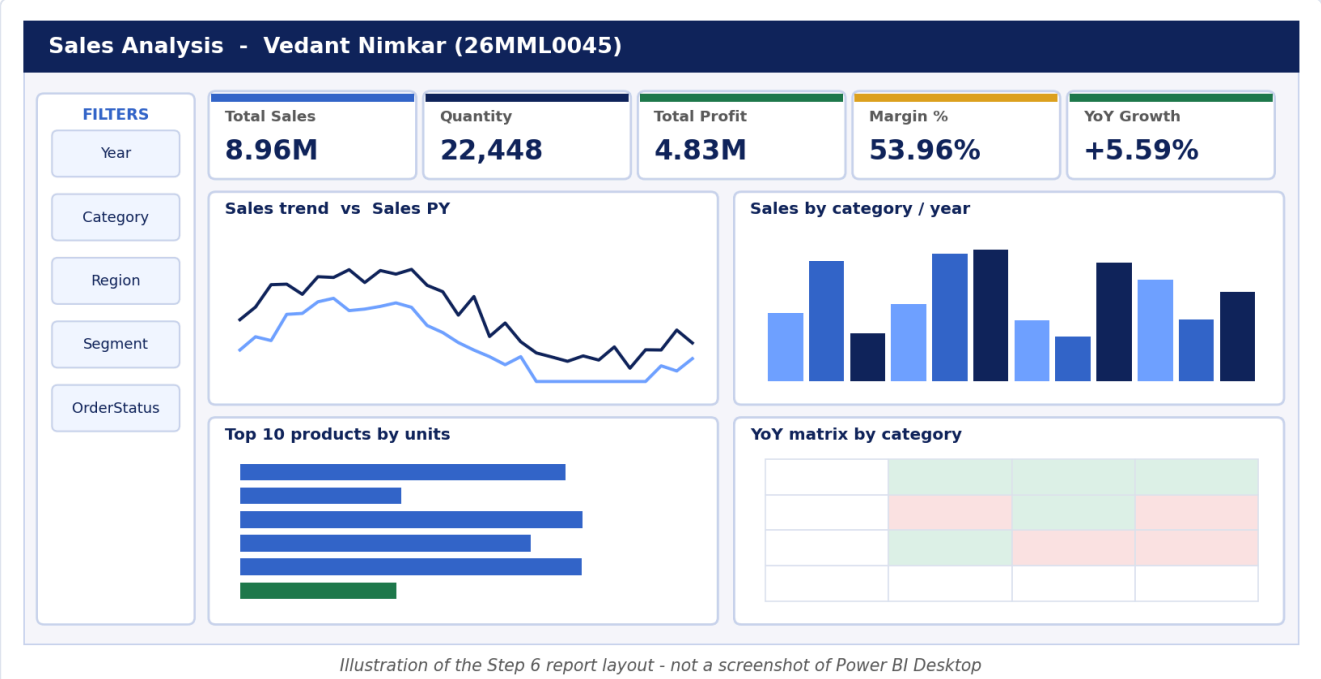


Figure 11 — Intended layout of the Page 1 report canvas (illustration of the design, not a Power BI screenshot)

Publishing procedure

1. **File → Save As →** `VisualAnalytics_Sales_26MML0045.pbix`
2. **Home → Publish**

3. Sign in with the institutional Microsoft 365 account
4. Select destination workspace → **My workspace**
5. On *Success!*, follow the link to open the report in the browser
6. **Dataset** → **Settings** → **Scheduled refresh** to automate refresh; configure an On-premises Data Gateway if the source is SQL Server

SCREENSHOT SLOT

Paste your Publish dialog and 'Success!' confirmation screenshots here

SCREENSHOT SLOT

Paste your published report rendered at app.powerbi.com here

Result

Both assessments were completed on a common star-schema sales dataset.

Assessment 1 — the source was connected and previewed; 6,012 raw rows were cleansed to 5,013 analysable rows by removing duplicates, null and negative quantities and cancelled orders, parsing three mixed date formats into a true date type and converting text currency to numeric; a star schema was then built with four active relationships, one deliberately inactive relationship, hidden surrogate keys, correct sort-by-column settings and four drill-down hierarchies.

Assessment 2 — calculated columns, twelve core measures and five calculated tables were authored in DAX, identifying `GlideMouse Wireless` as the top product by units (3,204) and `GamingBook X17` as the top product by revenue (2,534,432.17); `CALCULATE` was used with `ALL`, `ALLEXCEPT`, `ALLSELECTED`, `KEEPFILTERS` and `USERELATIONSHIP` to manipulate filter context; Time Intelligence functions produced YTD, prior-year, rolling and cumulative measures, yielding category YoY growth ranging from −1.13% to +66.45% against an overall +5.59%; a two-page report was then designed and published.

All measure values were cross-checked against an independent pandas computation over the same source data, recorded in `expected_results/EXPECTED_RESULTS.txt`.

Deliverables

Path	Contents
`data/`	CSV and Excel source files
`sql/01_create_load.sql`	SQL Server DDL and data load
`code/Step2_PowerQuery_M.txt`	all five Power Query M queries
`code/Step3_DataModel.md`	relationships, properties, hierarchies
`code/Step4_DAX_Calculated.txt`	calculated columns, measures, tables
`code/Step5_DAX_Advanced.txt`	CALCULATE and Time Intelligence
`code/Step6_Report_Design.md`	report layout and publish steps
`expected_results/`	verification values and cleaned reference data
`figures/`	figures reproduced in this record

> **Note on figures.** Figures 1, 2, 3, 4, 7, 8, 9 and 10 are rendered directly from the actual generated dataset and are therefore genuine results. Figures 5, 6 and 11 are drawn diagrams illustrating the intended model and report design. Screenshot slots are

provided throughout for the corresponding Power BI Desktop captures.